Chapter 5 Electrical physics

Section 5.1 Behaviour of charged particles

Worked example: Try yourself 5.1.1

AMOUNT OF CHARGE ON A GROUP OF ELECTRONS

Calculate the charge, in coulombs, carried by 4 million electrons.	
Thinking	**Working**
Express 4 million in scientific notation.	1 million = 10^6 4 million = 4×10^6
Calculate the charge, Q, in coulombs by multiplying the number of electrons by the charge on an electron (-1.6×10^{-19} C).	$q = (4 \times 10^6) \times (-e)$ $= (4 \times 10^6) \times (-1.6 \times 10^{-19}\text{ C})$ $= -6.4 \times 10^{-13}\text{ C}$

Worked example: Try yourself 5.1.2

NUMBER OF ELECTRONS IN A GIVEN AMOUNT OF CHARGE

The net charge on an object is $-4.8\,\mu$C ($1\,\mu$C = 1 microcoulomb = 10^{-6} C). Calculate the number of extra electrons on the object.	
Thinking	**Working**
Express $-4.8\,\mu$C in scientific notation.	$q = -4.8\,\mu\text{C}$ $= -4.8 \times 10^{-6}\text{ C}$
Find the number of electrons by dividing the charge on the object by the charge on an electron. (-1.6×10^{-19} C)	$n_e = \frac{q}{-e}$ $= \frac{-4.8 \times 10^{-6}\text{ C}}{-1.6 \times 10^{-19}\text{ C}}$ $= 3.0 \times 10^{13}$ electrons

Section 5.1 Review

KEY QUESTIONS SOLUTIONS

1 They will attract as they will be oppositely charged.

2 $n_e = \frac{-5.0\text{ C}}{-1.6 \times 10^{-19}\text{ C}} = 3.1 \times 10^{19}$

3 $q = 4.2 \times 10^{19} \times 1.6 \times 10^{-19}\text{ C} = +6.7\text{ C}$

4 Copper is a good conductor of electricity because its electrons are loosely held to their respective nuclei. This allows electrons to move freely through the material by ‘jumping’ from one atom to the next. Plastic is a good insulator. The plastic coating is used to insulate copper wiring to prevent charge leaving the circuit.

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Section 5.2 Energy in electric circuits

Worked example: Try yourself 5.2.1

USING THE DEFINITION OF POTENTIAL DIFFERENCE

A car battery can provide 3600 C of charge at 12 V. How much electrical potential energy is stored in the battery?	
Thinking	**Working**
Recall the definition of potential difference.	$\Delta V = \frac{E}{q}$
Rearrange this to make energy the subject.	$E = \Delta Vq$
Substitute in the appropriate values and solve.	$E = 12 \times 3600$ $= 43\,200\text{ J}$ $= 4.3 \times 10^4\text{ J}$

Section 5.2 Review

KEY QUESTIONS SOLUTIONS

1 A. When a conductor links two bodies between which there is a potential difference, charges will flow through the conductor until the potential difference is equal to zero.

2 $\Delta V = \frac{E}{q}$

a $\frac{40}{10} = 4\text{ V}$

b $\frac{15}{10} = 1.5\text{ V}$

c $\frac{20}{10} = 2\text{ V}$

3 $\Delta V = \frac{E}{q}$

$= \frac{100}{5}$

$= 20\text{ V}$

4 $E = \Delta Vq$

$2 \times 10^3\text{ J} = q(12\text{ V})$

$q = 167\text{ C}$

5 **a** heat and light

b $E = \Delta Vq$

$q = \frac{E}{V}$

$= \frac{(3.6 \times 10^3)}{(240)}$

$= 15\text{ C}$

6 the gravitational potential energy of the water

7 The voltmeter must always be in parallel with the light bulb, i.e. at M2 or M3.

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Section 5.3 Electric current and circuits

Worked example: Try yourself 5.3.1

USING $I = \frac{q}{t}$

Calculate the number of electrons that flow past a particular point each second in a circuit that carries a current of 0.75 A.	
Thinking	**Working**
Rearrange the equation $I = \frac{q}{t}$ to make q the subject.	$I = \frac{q}{t}$ so $q = I \times t$
Calculate the amount of charge that flows past the point in question by substituting the values given.	$q = 0.75 \times 1$ $= 0.75\,\text{C}$
Find the number of electrons by dividing the charge by the charge on an electron (1.6×10^{-19} C).	$n_e = \frac{q}{q_e}$ $= \frac{0.75}{1.6 \times 10^{-19}}$ $= 4.69 \times 10^{18}$ electrons

Worked example: Try yourself 5.3.2

USING $E = \Delta VIt$

A potential difference of 12 V is used to generate a current of 1750 mA to heat water for 7.5 minutes. Calculate the energy transferred to the water in that time.	
Thinking	**Working**
Convert quantities to SI units.	$\frac{1750\ \text{mA}}{100} = 1.75\,\text{A}$ $7.5\,\text{min} \times 60\,\text{s} = 450\,\text{s}$
Substitute values into the equation and calculate the amount of energy in joules.	$E = \Delta VIt$ $= 12 \times 1.750 \times 7.5 \times 60$ $= 9.45 \times 10^3\,\text{J}$

Worked example: Try yourself 5.3.3

USING $P = \Delta VI$

An appliance running on 120 V draws a current of 6 A. Calculate the power used by this appliance.	
Thinking	**Working**
Identify the relationship needed to solve the problem.	$P = \Delta VI$
Identify the required values from the question, substitute and calculate.	$P = 120 \times 6$ $= 720\,\text{W}$

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Section 5.3 Review

KEY QUESTIONS SOLUTIONS

1 A continuous conducting loop (closed circuit) must be created from one terminal of a power supply to the other terminal.

2 Cell, light bulb, open switch, resistor and ammeter.

3 C. It's now known that charge carriers are electrons, which flow from the negative terminal to the positive terminal of the battery.

4 $I = \frac{q}{t}$ in coulombs and seconds

a 3 A

b 0.5 A

c 0.008 A

5 Use $q = It$, with I in amperes and t in seconds.

a 5 C

b 300 C

c 18 000 C

6 **a** $q = It = (5 \times 10^{-3}) \times (600) = 3\text{ C}$

b $q = 200 \times 5 = 1000\text{ C}$

c $q = (400 \times 10^{-3}) \times (3600) = 1440\text{ C}$

7 **a** $q = n_e \times q_e$

$= 10^{20} \times 1.6 \times 10^{-19}\text{ C}$

$= 16\text{ C}$

b $I = \frac{q}{t}$

$= \frac{16}{4}$

$= 4\text{ A}$

8 3.2 C flow past a point in 10 seconds. Calculate:

a $n_e = \frac{q}{q_e}$

$= \frac{3.2}{1.6 \times 10^{-19}}$

$= 2 \times 10^{19}$ electrons

b $I = \frac{q}{t}$

$= \frac{3.2}{10}$

$= 0.32\text{ A}$

9 B. Circuit B is the only circuit in which the current passes through both globes and the ammeter is in series with both globes when the switch is closed.

10 **a** $t = 5 \times 60 = 300\text{ s}$

$E = P \times t$

$= 460 \times 300$

$= 138\ 000\text{ J}$ (or 138 kJ)

b $I = \frac{P}{V}$

$= \frac{460}{230}$

$= 2\text{ A}$

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Section 5.4 Resistance

Worked example: Try yourself 5.4.1

USING OHM'S LAW TO CALCULATE RESISTANCE

An electric bar heater draws 10 A of current when connected to a 240 V power supply. Calculate the resistance of the element in the heater.

Thinking	Working
Ohm's law is used to calculate resistance.	$\Delta V = IR$
Rearrange the equation to find R.	$R = \frac{V}{I}$
Substitute in the known values.	$R = \frac{240}{10}$ $= 24\,\Omega$

Worked example: Try yourself 5.4.2

USING OHM'S LAW TO CALCULATE RESISTANCE, CURRENT AND POTENTIAL DIFFERENCE

The table below shows measurements for the potential difference and corresponding current for an ohmic conductor.

ΔV (V)	0	3	9	ΔV_2
I (A)	0	0.20	I_1	0.80

Determine the missing results, I_1 and V_2.

Thinking	Working
Determine the factor by which potential difference has increased from the second column to the third column.	$\frac{9}{3} = 3$ The potential difference has tripled.
Apply the same factor increase to the current in the second column, to determine the current in the third column (I_1).	$I_1 = 3 \times 0.20$ $= 0.6\,A$
Determine the factor by which current has increased from the second column to the fourth column.	$\frac{0.80}{0.20} = 4$ The current has increased by a factor of 4.
Apply the same factor increase to the potential difference in the second column, to determine the potential difference in the fourth column (ΔV_2).	$\Delta V_2 = 4 \times 3$ $= 12\,V$

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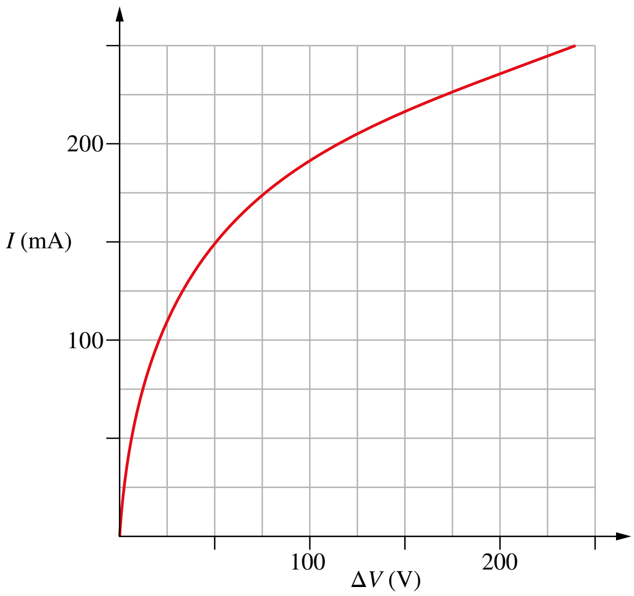
Worked example: Try yourself 5.4.3

CALCULATING RESISTANCE FOR A NON-OHMIC CONDUCTOR

A 240 V, 60 W incandescent light globe has the I–ΔV characteristics shown in the graph. Calculate the resistance of the light globe when the potential difference is 175 V.



Thinking	Working
From the graph, determine the current at the required potential difference. Note that current is given in mA, so convert it to A.	At $\Delta V = 175$ V, $I = 225$ mA. Therefore $I = 0.225$ A
Substitute these values into Ohm's law to find the resistance.	$R = \frac{V}{I}$ $= \frac{175}{0.225}$ $= 778\,\Omega$

Worked example: Try yourself 5.4.4

USING OHM'S LAW TO FIND CURRENT

The element of a bar heater has a resistance of 25 Ω. Calculate the current (in mA) that will flow through this element if it is connected to a 240 V supply.

Thinking	Working
Recall Ohm's law.	$\Delta V = IR$
Rearrange the equation to make I the subject.	$I = \frac{V}{R}$
Substitute in the values for this problem and solve.	$I = \frac{240}{25}$ $= 9.60$ A
Convert the answer to the required units.	$I = 9.6$ A $= 9.60 \times 10^3$ mA $= 9600$ mA

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Worked example: Try yourself 5.4.5

USING OHM'S LAW TO FIND POTENTIAL DIFFERENCE

The globe of a torch has a resistance of 5.7 Ω when it draws 700 mA of current. Calculate the potential difference across the globe.	
Thinking	**Working**
Convert 700 mA to A.	$700 \times 10^{-3} = 0.7\,\text{A}$
Recall Ohm's law.	$\Delta V = IR$
Substitute in the known values and solve.	$\Delta V = 0.7 \times 5.7$ $= 3.99\,\text{V}$

Section 5.4 Review

KEY QUESTIONS SOLUTIONS

1 **a** A, B, C
b C, B, A

2
$$R = \frac{V}{I} = \frac{2}{0.25} = 8\,\Omega$$
$$I = \frac{V}{R}$$
$$I_1 = \frac{3}{8} = 0.375\,\text{A}$$
$$\Delta V = IR$$
$$\Delta V_2 = 0.60 \times 8 = 4.8\,\text{V}$$

3 **a** The wire is ohmic. This is because there is a proportional relationship between the voltage and the current, as shown by the linear nature of the I–ΔV graph, which means that the resistance is a constant.
b 3 A
c
$$R = \frac{V}{I} = \frac{25}{10} = 2.5\,\Omega$$

4 **a**
$$R = \frac{V}{I} = \frac{2.5}{3.5} = 0.71\,\Omega$$
b Since $V = IR$, if we double our potential difference, we would expect the current to double if the resistance of the ohmic resistor is constant. Therefore:
$2V = (2 \times I)R$
$(2 \times 2.5) = (2 \times 3.5)R$
$5 = 7R$
And we can see the value of R has remained constant.

5 They are both right. The resistance of the device is different for different voltages. Therefore the device is non-ohmic.

6
$$R = \frac{5}{45 \times 10^{-3}} = 111.11\,\Omega$$
$$I = \frac{8}{111.11} = 72\,\text{mA}$$

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7 **a** $R = \frac{V}{I} = \frac{4}{2} = 2\,\Omega$

b $I = \frac{10}{2} = 5\,\text{A}$

8 **a** $R = \frac{kL}{A}$

$0.8 = \frac{kL}{A}$ then $1.6 = \frac{k(2L)}{A} = 1.6\,\Omega$

b A wire of twice the diameter has four times the cross-sectional area.

Then $R = \frac{0.8}{4} = 0.2\,\Omega$.

9 **a** It is non-ohmic, as the I–ΔV relationship is nonlinear.

b From the graph, when $\Delta V = 10\,\text{V}$, $I = 0.5\,\text{A}$.

c For $I = 1.0\,\text{A}$, $\Delta V = 15\,\text{V}$.

d The resistance of the device at these voltages will be given by $R = \frac{V}{I}$.

i For 10 V, $R = \frac{10}{0.5} = 20\,\Omega$.

ii For 20 V, $R = \frac{20}{1.5} = 13.3\,\Omega$.

Section 5.5 Series and parallel circuits

Worked example: Try yourself 5.5.1

CALCULATING AN EQUIVALENT SERIES RESISTANCE

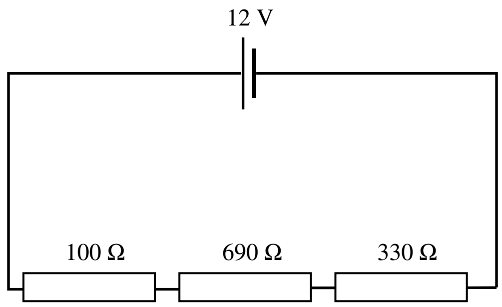
A string of Christmas lights consists of 20 light bulbs connected in series. Each bulb has a resistance of 8 Ω. Calculate the equivalent series resistance of the Christmas lights.

Thinking	Working
Recall the formula for equivalent series resistance.	$R_T = R_1 + R_2 + \ldots + R_n$
Substitute in the given values for resistance. As there are 20 equal globes, you can multiply 8 Ω by 20 globes.	$R_T = 20 \times 8$ $= 160\,\Omega$

Worked example: Try yourself 5.5.2

USING EQUIVALENT SERIES RESISTANCE FOR CIRCUIT ANALYSIS

Use an equivalent series resistance to calculate the current flowing in the series circuit below and the potential difference across each resistor.



Thinking	Working
Recall the formula for equivalent series resistance.	$R_T = R_1 + R_2 + R_3 + \ldots + R_n$
Find the equivalent (total) resistance in the circuit.	$R_T = 100 + 690 + 330 = 1120\,\Omega$

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Use Ohm's law to calculate the current in the circuit. Whenever calculating current in a series circuit, use R_T and the potential difference of the power supply.	$I = \frac{V}{R}$ $= \frac{12}{1120}$ $= 0.011\text{ A}$
Use Ohm's law to calculate the potential difference across each separate resistor.	$\Delta V = IR$ Therefore: $\Delta V_1 = 0.011\text{ A} \times 100\,\Omega = 1.1\text{ V}$ $\Delta V_2 = 0.011\text{ A} \times 690\,\Omega = 7.6\text{ V}$ $\Delta V_3 = 0.011\text{ A} \times 330\,\Omega = 3.6\text{ V}$
Use the loop rule to check the answer.	$\Delta V_T = \Delta V_1 + \Delta V_2 + \Delta V_3$ $= 1.1 + 7.6 + 3.6$ $= 12.3\text{ V}$ Since this is approximately the same as the potential difference provided by the cell, the answer is reasonable.

Worked example: Try yourself 5.5.3

CALCULATING AN EQUIVALENT PARALLEL RESISTANCE

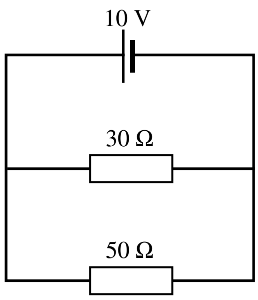
A $20\,\Omega$ resistor is connected in parallel with a $50\,\Omega$ resistor. Calculate the equivalent parallel resistance.

Thinking	**Working**
Recall the formula for equivalent effective resistance.	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$
Substitute in the given values for resistance.	$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{50}$
Solve for R_T.	$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{50}$ $= \frac{5}{100} + \frac{2}{100}$ $= \frac{7}{100}$ $R_T = \frac{100}{7}$ $= 14.3\,\Omega$

Worked example: Try yourself 5.5.4

USING EQUIVALENT PARALLEL RESISTANCE FOR CIRCUIT ANALYSIS

Find the equivalent parallel resistance to calculate the current flowing out of the 10 V cell in the parallel circuit shown. Then find the current flowing through each resistor.



Thinking	**Working**
Recall the formula for equivalent parallel resistance.	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$

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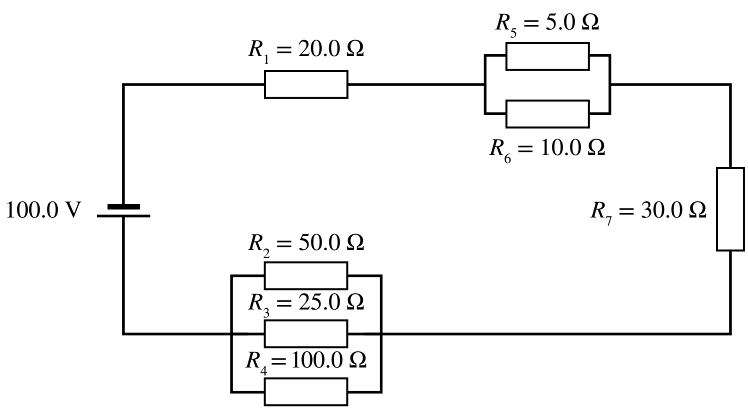
Substitute in the given values for resistance.	$\frac{1}{R_T} = \frac{1}{30} + \frac{1}{50}$
Solve for R_T.	$\frac{1}{R_T} = \frac{1}{30} + \frac{1}{50}$ $\frac{1}{R_T} = \frac{5}{150} + \frac{3}{150}$ $\frac{1}{R_T} = \frac{8}{150}$ $R_T = \frac{150}{8}$ $= 18.8\,\Omega$
Use Ohm's law to calculate the current in the circuit. To calculate I, use the potential difference of the power supply and the total resistance.	$I_{circuit} = \frac{V}{R} = \frac{10}{18.8} = 0.53\,A$
Use Ohm's law to calculate the current through each resistor. Remember that the potential difference across each resistor is the same as the potential difference of the power supply, 10 V in this case.	$30\,\Omega$ resistor: $I_{30} = \frac{V}{R} = \frac{10}{30} = 0.33\,A$ $50\,\Omega$ resistor: $I_{50} = \frac{V}{R} = \frac{10}{50} = 0.20\,A$
Use the junction rule to check the answers.	$I_{circuit} = I_{30} + I_{50}$ $0.53\,A = 0.33\,A + 0.20\,A$ This is correct, so the answers are reasonable.

Worked example: Try yourself 5.5.5

COMPLEX CIRCUIT ANALYSIS

Calculate the potential difference across and the current through each resistor in the circuit below.



Thinking	Working
Find an equivalent resistance for the two parallel resistors. The effective resistance of these should be less than the smaller resistor, that is, less than $5.0\,\Omega$.	$\frac{1}{R_{5-6}} = \frac{1}{R_5} + \frac{1}{R_6}$ $= \frac{1}{5.0} + \frac{1}{10.0}$ $= \frac{2}{10} + \frac{1}{10}$ $R_{2-3} = \frac{10}{3} = 3.33\,\Omega$
Find an equivalent resistance for the three parallel resistors. The effective resistance of these should be less than the smaller resistor, that is, less than $5.0\,\Omega$.	$\frac{1}{R_{2,3,4}} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$ $= \frac{1}{50} + \frac{1}{25} + \frac{1}{100}$ $= \frac{2}{100} + \frac{4}{100} + \frac{1}{100}$ $R_{2,3,4} = \frac{100}{7} = 14.3\,\Omega$

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Find an equivalent series resistance for the circuit as the circuit can now be thought of as four resistors in series: 20.0 Ω, 14.3 Ω, 3.33 Ω and 30.0 Ω.	$R_T = 20.0\,\Omega + 14.3\,\Omega + 3.33\,\Omega + 30.0\,\Omega$ $= 67.63\,\Omega$
Use Ohm's law to calculate the current in the circuit. Use the potential difference of the power supply and the total resistance to do this calculation.	$\Delta V = IR$ $I = \frac{V}{R}$ $= \frac{100.0}{67.6}$ $= 1.48\,\text{A}$
Use Ohm's law to calculate the potential difference across each resistor (or parallel group of resistors) in series. (Note that the potential difference across R_2 is the same as that across R_3 as they are in parallel.)	$\Delta V = IR$ $\Delta V_1 = 1.48 \times 20.0 = 29.6\,\text{V}$ $\Delta V_{2,3,4} = 1.48 \times 14.3 = 21.2\,\text{V}$ $\Delta V_{5,6} = 1.48 \times 3.33 = 4.93\,\text{V}$ $\Delta V_7 = 1.48 \times 30.0 = 44.4\,\text{V}$ Check: 29.6 + 21.2 + 4.93 + 44.4 = 100.13 V (with some slight rounding error = 100) This confirms that the loop rule holds for this circuit.
Use Ohm's law where necessary to calculate the current through each resistor.	$I_1 = I_7 = 1.48\,\text{A}$ $I = \frac{V}{R}$ $I_2 = \frac{21.2}{50.0} = 0.424\,\text{A}$ $I_3 = \frac{21.2}{25.0} = 0.848\,\text{A}$ $I_4 = \frac{21.2}{100} = 0.212\,\text{A}$ Check: 0.424 + 0.848 + 0.212 = 1.48 A (with some slight rounding error) This confirms that the junction rule holds for this parallel section. $I_5 = \frac{4.93}{5.0} = 0.986\,\text{A}$ $I_6 = \frac{4.93}{10.0} = 0.493\,\text{A}$ Check: 0.968 + 0.493 = 1.48 (with some slight rounding error) This confirms that the junction rule holds for this parallel section.

Worked example: Try yourself 5.5.6

COMPARING POWER IN SERIES AND PARALLEL CIRCUITS

Consider a 200 Ω and an 800 Ω resistor wired in parallel with a 12 V cell. Calculate the power drawn by these resistors. Compare this to the power drawn by the same two resistors when wired in series.	
Thinking	**Working**
Calculate the equivalent resistance for the parallel circuit.	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$ $= \frac{1}{200} + \frac{1}{800}$ $= \frac{4}{800} + \frac{1}{800}$ $= \frac{5}{800}$ $R_T = \frac{800}{5}$ $= 160\,\Omega$

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Calculate the total current drawn by the parallel circuit.	$\Delta V = IR$ $I = \frac{V}{R} = \frac{12}{160} = 0.075\,\text{A}$
Use the power equation to calculate the power drawn by the parallel circuit.	$P = \Delta VI$ $= 12 \times 0.075$ $= 0.90\,\text{W}$
Calculate the equivalent resistance for the series circuit.	$R_T = R_1 + R_2 + \ldots + R_n$ $= 200 + 800$ $= 1000\,\Omega$
Calculate the total current drawn by the series circuit.	$\Delta V = IR$ $I = \frac{12}{1000} = 0.012\,A$
Use the power equation to calculate the power drawn by the series circuit.	$P = \Delta VI$ $= 12 \times 0.012$ $= 0.144\,\text{W}$
Compare the power drawn by the two circuits.	$\frac{P_{\text{parallel}}}{P_{\text{series}}} = \frac{0.90}{0.144} = 6.25$ The parallel circuit draws over 6 times as much power as the series circuit.

Section 5.5 Review

KEY QUESTIONS SOLUTIONS

1 B

$R_T = R_1 + R_2$

$= 20 + 20 = 40\,\Omega$

$1 = \frac{V}{R_1}$

$= \frac{6}{40} = 0.15\,\text{A}$

ΔV across each resistor:

$\Delta V = IR = 0.15 \times 20 = 3\,\text{V}$

(Or, as the resistors are equal, the same voltage will be lost across each and will add to 6 V, so 3 V must be lost across each resistor.)

2 **a** $R_T = R_1 + R_2 + R_3$

$= 100 + 250 + 50 = 400\,\Omega$

$I_T = \frac{V}{R_T} = \frac{3}{400} = 0.0075\,\text{A}$ or $7.5\,\text{mA}$

b $R = 100\,\Omega$ and $I = 0.0075\,\text{A}$

$\Delta V_{100} = IR$

$= 0.0075 \times 100 = 0.75\,\text{V}$

3 $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_1} + \frac{1}{R_1}$ (the resistors are identical, so $R_1 = R_2$)

$= \frac{2}{R_T}$

$\therefore R_T = \frac{R_1}{2}$

$R_1 = 2 \times R_T = 2 \times 68 = 136\,\Omega$

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4 **a** $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$

$= \frac{1}{20} + \frac{1}{10}$

$= \frac{1}{20} + \frac{2}{20}$

$= \frac{3}{20}$

$R_T = \frac{20}{3}$

$= 6.67\,\Omega$

$I = \frac{\Delta V}{R}$

$I_T = \frac{5}{6.67}$

$= 0.75\,\text{A}$

b $I_{20} = \frac{V_{20}}{R} = \frac{5}{20} = 0.25\,\text{A}$

c $I_{20} = \frac{V_{10}}{R} = \frac{5}{10} = 0.5\,\text{A}$

5 **a** $\Delta V = IR$

$\Delta V_{40} = 0.3 \times 40 = 12\,\text{V}$ (300 mA = 0.3 A)

Since the components are in parallel the voltage across the 40 Ω resistor (or the 60 Ω resistor) is also the voltage of the battery.

b $I_{60} = \frac{V_{60}}{R} = \frac{12}{60} = 0.2\,\text{A}$ (or 200 mA)

6 First determine the total resistance of the circuit:

$\frac{1}{R_{3-4}} = \frac{1}{R_3} + \frac{1}{R_4}$

$= \frac{1}{10} + \frac{1}{10}$

$R_{3-4} = 5\,\Omega$

$R_T = 20 + 15 + 5 = 40\,\Omega$

$I_T = \frac{V_T}{R_T} = \frac{12}{40} = 0.3\,\text{A}$ (or 300 mA)

$I_1 = I_2 = I_T = 0.3\,\text{A}$ (since these are in series)

$\Delta V_1 = I_1 R_1 = 0.3 \times 20 = 6\,\text{V}$

$\Delta V_2 = I_2 R_2 = 0.3 \times 15 = 4.5\,\text{V}$

$\Delta V_3 = \Delta V_4 = I_{3-4} R_{3-4} = 0.3 \times 5 = 1.5\,\text{V}$

$I_3 = I_4 = \frac{V_3}{R_3} = \frac{V_4}{R_4} = \frac{1.5}{10} = 0.15\,\text{A}$

7 $R_{\text{top row}} = 3 + 4 = 7\,\Omega$

$R_{\text{bottom row}} = 5 + 6 = 11\,\Omega$

$\frac{1}{R_{\text{parallel}}} = \frac{1}{7} + \frac{1}{11} = \frac{11}{77} + \frac{7}{77} = \frac{18}{77}$

$R_{\text{parallel}} = 4.278\,\Omega$

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8 **a** $R_T = R_1 + R_2 + \ldots + R_n$

$= 20 + 20 + 20 + 20$

$= 80\,\Omega$

$\Delta V = IR$

$\therefore I = \frac{V}{R}$

$= \frac{10}{80}$

$= 0.125\,\text{A}$

$P = \Delta VI$

$= 10 \times 0.125$

$= 1.25\,\text{W}$

b $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \ldots + \frac{1}{R_n}$

$= \frac{1}{20} + \frac{1}{20} + \frac{1}{20} + \frac{1}{20}$

$= \frac{4}{20}$

$\frac{R_T}{1} = \frac{20}{4}$

$R_T = 5\,\Omega$

$\Delta V = IR$

$\therefore I = \frac{V}{R} = \frac{10}{5}$

$= 2\,\text{A}$

$P = \Delta VI$

$= 10 \times 2$

$= 20\,\text{W}$

9 C. Parallel wiring allows each appliance to be switched on and off independently (and also receive mains voltage supply).

Section 5.6 Electrical safety

Worked example: Try yourself 5.6.1

CALCULATING THE COST OF ELECTRICITY

A 2500 W iron is used for 2.5 hours. Assume the price for household electricity is 26 cents per kWh. How much would it cost (to the nearest cent) to use this iron for 2.5 hours?	
Thinking	**Working**
Convert the power consumption of the appliance to kW.	$\frac{2500}{1000} = 2.5$ $= 2.5\,\text{kW}$
Use the appropriate equation to multiply the power of the appliance in kW by the number of hours it operates.	$E = Pt$ $= 2.5 \times 2.5$ $= 6.25\,\text{kWh}$
Multiply the number of kWh by the cost per kWh.	Cost $= 6.25 \times 0.26$ $= \$1.63$ (to two decimal places)

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Section 5.6 Review

KEY QUESTIONS SOLUTIONS

1 A. In the event of an electrical fault the current will rapidly increase through the zero resistance path offered by the earth connection. Once this current exceeds the fuse's rating the fuse will blow, shutting off power to the appliance.

2 D. Double insulated appliances usually do not have an earth connection.

3. $1\,\text{kWh} = 1000\,\text{W} \times 3600\,\text{s}$

$= 3\,600\,000\,\text{J}$

$10\,\text{kWh} = 10 \times 3\,600\,000$

$= 3.6 \times 10^7\,\text{J}$

4 This air conditioner would cost $0.75 \times 5 \times 0.27$ = approximately \$1 to run for 5 hours. Therefore, the figure \$10 in the statement is incorrect.

5 The neutral and earth are common.

6 It is much safer to place the fuse in the active circuit because then it cuts off the supply to the circuit.

7 The earth stake ensures that the neutral and earth conductors are at zero potential.

8 The toaster will work normally, but the connection is very unsafe because it will remain live even when switched off.

9 The outer casing of the appliance could become live.

10 $I = \dfrac{240\,\text{V}}{1.0 \times 10^5} = 2.4\,\text{mA}$

CHAPTER 5 REVIEW

1 $n_e = \dfrac{q}{q_e}$

$= \dfrac{-3}{-1.6 \times 10^{-19}}$

$= 1.9 \times 10^{19}$ electrons

2 $q = n_e \times q_e$

$= 4.2 \times 10^{19} \times 1.6 \times 10^{-19}\,\text{C}$

$= 6.7\,\text{C}$

3 A. In a solid metal, electrons are the only charged particles that are free to move. Electrons are negatively charged.

4 $q = n_e \times q_e$

$= 2 \times 1.6 \times 10^{-19}\,\text{C}$

$= 3.2 \times 10^{-19}\,\text{C}$

5 $I = \dfrac{q}{t}$

$= \dfrac{0.23}{60}$

$= 3.8 \times 10^{-3}\,\text{A}$

6 Conventional current represents the flow of charge around a circuit as if the moving charges were positive, which means the direction is from the positive terminal to the negative terminal. In reality, the moving particles in a metal wire are negatively charged electrons. Electron flow describes the movement of these electrons from the negative terminal to the positive terminal.

7 **a** $q = It$

$= 1.6 \times 100$

$= 160\,\text{C}$

b $n_e = \dfrac{q}{q_e}$

$= \dfrac{160}{1.6 \times 10^{-19}}$

$= 10^{21}$ electrons

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8 **a** $q = n_e \times q_e$

$= 5 \times 10^{18} \times 1.6 \times 10^{-19}$

$= 0.8\,C$

b $t = \frac{q}{I}$

$= \frac{0.8}{0.04}$

$= 20$ seconds

9 $E = \Delta Vq$

$= 3.8 \times 2$

$= 7.6\,J$

10 $V = \frac{E}{q}$

$= \frac{2}{0.5}$

$= 4\,V$

11 $P = \frac{E}{t}$

$= \frac{2500}{30 \times 60}$

$= 1.39\ W$

12 $V = \frac{E}{q}$

$= \frac{1.4 \times 10^{-18}}{1.6 \times 10^{-19}}$

$= 8.75\,V$

13 $I = \frac{P}{V}$

$= \frac{2000}{230}$

$= 8.7\,A$

14 At 50 V, $I = 150\,mA = 0.15\,A$

$R = \frac{V}{I}$

$= \frac{50}{0.15}$

$= 333\,\Omega$

15 A. The equivalent resistance of resistors in series is the sum of their individual resistances.

16 **a** $R_T = R_{\text{parallel pair}} + R_3$

$\therefore R_3 = R_T - R_{\text{parallel pair}} = 8.5 - 5 = 3.5\,\Omega$

b $I_3 = I_T = \frac{V_T}{R_T} = \frac{3}{8.5} = 0.35\,A$

c $\Delta V_3 = I_3 \times R_3 = 0.35 \times 3.5 = 1.2\,V$

$\Delta V_{\text{parallel pair}} = 3 - 1.2 = 1.8\,V$

d $I_2 = \frac{V_2}{R_2} = \frac{1.8}{15} = 0.12\,A$

e $I_1 = I_T - I_2 = 0.35 - 0.12 = 0.23\,A$

f $R_1 = \frac{V_1}{I_1} = \frac{1.8}{0.23} = 7.83\,\Omega$

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17 **a** Ammeter. The meter is connected in series so it must be an ammeter.

b $\frac{1}{R_{\text{top parallel group}}} = \frac{1}{40} + \frac{1}{40} = \frac{2}{40}$

$R_{\text{top parallel group}} = 20\,\Omega$

$\frac{1}{R_{\text{bottom parallel group}}} = \frac{1}{20} + \frac{1}{60} = \frac{3}{60} + \frac{1}{60} = \frac{4}{60}$

$R_{\text{bottom parallel group}} = 15\,\Omega$

$\frac{1}{R_{\text{total}}} = \frac{1}{20} + \frac{1}{15}$

$= \frac{3}{60} + \frac{4}{60} = \frac{7}{60}$

$R_{\text{total}} = \frac{60}{7} = 8.57\,\Omega$

18 The earth wire is usually connected to the metal casing of an electrical appliance. If the insulation around the wire inside the appliance becomes degraded, the casing of the appliance could become live and dangerous to touch. In this situation, the earth wire provides an alternative low-resistance path to earth, protecting users of the appliance from electrocution.

19 The circuit will need to have either two pairs of series resistors connected in parallel or two pairs of parallel resistors connected in series.

20 **a** $R_T = R_1 + R_2 + R_3 = 20 + 20 + 20 = 60\,\Omega$

$\Delta V = IR$

$\therefore I = \frac{V}{R} = \frac{12}{60} = 0.2\,\text{A}$

$P = \Delta VI = 12 \times 0.2 = 2.4\,\text{W}$

b $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{20} + \frac{1}{20} + \frac{1}{20} = \frac{3}{20}$

$R_T = \frac{20}{3} = 6.67\,\Omega$

$I = \frac{V}{R} = \frac{12}{6.67} = 1.8\,\text{A}$

$P = \Delta VI = 12 \times 1.8 = 21.6\,\text{W}$

c $\frac{P_{\text{parallel}}}{P_{\text{series}}} = \frac{21.6}{2.4} = 9$

The parallel circuit draws 9 times more power.

21 D. A 50 mA current for over 4.5 s is likely to cause severe shock and possible death.

22 $E = Pt$

$= 3 \times 4$

$= 12\,\text{kWh}$

Cost $= 12 \times 0.30$

$= \$3.60$

23 D. Power is a measure of how quickly energy is consumed/supplied/transformed.

1 watt = 1 joule per second

24 **a** $\frac{1}{R_T} = \frac{1}{100} + \frac{1}{200} + \frac{1}{600}$

$= \frac{6}{600} + \frac{3}{600} + \frac{1}{600}$

$= \frac{10}{600}$

$R_T = \frac{600}{10} = 60\,\Omega$

b $\Delta V = IR$

$120 = I \times 60$

$I = 2.0\,\text{A}$

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c Each branch has 120 V across it

$I_1 = \frac{V}{R_T} = \frac{120}{100} = 1.2\,A$

$I_2 = \frac{V}{R_T} = \frac{120}{200} = 0.6\,A$

$I_3 = \frac{V}{R_T} = \frac{120}{600} = 0.2\,A$

$I_1 = 1.20\,A$, $I_2 = 0.60\,A$, $I_3 = 0.20\,A$

d $P = \Delta VI$

$= 120 \times (1.2 + 0.6 + 0.2)$

$= 240\,W$

e 240 W

25 **a** 4 V. This is calculated using Ohm's law, or alternatively, by recognising that the voltage drop at ΔV_{out} is half the voltage drop across the LDR.

$\Delta V = IR$

$12 = I \times 300$

$I = 0.04\,A$

$\Delta V = IR$

$= 0.04 \times 100$

$= 4\,V$

b above; as light increases, the resistance of the LDR decreases, hence V_{out} rises

c V_{out} approaches zero, as the LDR has increased resistance and therefore the voltage drop across the LDR approaches 12 V.

26 **a** Combining Ohm's law, $\Delta V = IR$, and the equation for power:

$P = \Delta VI = I^2R = \frac{V^2}{R}$

$25 = I_X^2\, R_X$

$I_X^2 = \frac{25}{100} = 25 \times 10^{-2}$

$I_X = 5 \times 10^{-1} = 0.5\,A$

$\Delta V_X = IR_X = 0.5 \times 100 = 50\,V$

$\Delta V_Y = R_Y \times \frac{1}{2} \times 0.5 = 100 \times 0.25 = 25\,V$

$\Delta V_{total} = \Delta V_X + \Delta V_Y = 75\,V$

b $P = \Delta VI = 75 \times 0.5 = 37.5\,W$

27 **a** 3

b 1

c 2

28 The finger provides less contact with the live wire and hence more resistance.

29 A fuse will melt when a high current flows in a circuit. Without the fuse the heat generated from a high current could be enough to start a fire and burn the house down. A safety switch switches off a circuit when the current in the active and neutral wires are not equal, thus preventing possible electrocution.

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